

Updates to the MSFC Meteoroid Stream Model

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Abstract The Marshall Space Flight Center (MSFC) Meteoroid Stream Model simulates particle ejection and subsequent evolution from comets in order to provide meteor shower forecasts to spacecraft operators for hazard mitigation and planning purposes. The model, previously detailed in Moser and Cooke (Earth Moon Planets **95**, 141 (2004)), has recently been updated; the changes include the implementation of the RADAU integrator, an improved planetary treatment, and the inclusion of general relativistic effects in the force function. The results of these updates are investigated with respect to various meteoroid streams and the outcome presented.

Keywords Aurigids · Comet ejection · Draconids · Leonids · Meteor shower · Meteoroids · Model predictions · Numerical integration · Perseids · Stream model

1 Introduction

The NASA Meteoroid Environment Office has developed the Marshall Space Flight Center (MSFC) Meteoroid Stream Model to forecast meteor showers for Earth and Earth-orbiting spacecraft to provide information to spacecraft operators for hazard mitigation and mission planning. Changes to the model, previously presented in Moser and Cooke (2004), have recently been implemented. The updates include the use of a new numerical integrator, the inclusion of more planetary effects, and improvements in the calculation of planetary positions. General relativistic effects are also now taken into account. The immediate aim of this paper is to investigate the effect these updates had on various Leonid and Perseid

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streams in regards to peak time and duration, and to show the results of modeling the October Draconids and α -Aurigids with the model for the first time.

2 Model

2.1 Overview

In modeling particle ejection and subsequent evolution from comets, the workload is broken into three parts. First the test particles are created for each cometary perihelion passage, then their positions and velocities are integrated forward in time, and finally the particles are examined at specific times of interest. The first and third steps are detailed in Moser and Cooke (2004). It is the second step that will be discussed here as it is the one that has been affected by the recent update.

2.2 Updates

In the previous version of the MSFC Meteoroid Stream Model, a 4th order variable step-size Runge-Kutta (RK4) integrator was used to integrate meteoroid position and velocity forward in time. In this update, a 15th order RADAU integrator (Everhart 1985) has replaced the RK4. It is more accurate than the RK4, especially when close planetary approaches must be considered. Also of note is that it has been used successfully to determine the orbits of over 200 comets (Marsden et al. 1978; Everhart and Marsden 1983) and it is used by other stream modelers with good results (i.e. McNaught and Asher 2001; Vaubaillon 2002).

In the original model, the effects of radiation pressure, Poynting-Robertson drag, and the gravitational influences of 7 planets, Venus through Neptune, were taken into account. Mercury's mass was included in the mass of the Sun, and perturbations from the Earth-Moon barycenter were included, instead of treating the Earth and Moon separately. Jet Propulsion Laboratory's (JPL) DE406 (Standish 1998) was used to compute the positions of the planets: planetary positions were interpolated with a cubic spline subroutine from a look-up table of positions given every day from 1000 CE to 2150 CE.

For more accurate orbits, radiation pressure, Poynting-Robertson drag, and the gravitational influences of 8 planets, Mercury through Neptune, and 2 minor bodies, the Moon (treated separately from the Earth) and Pluto, are considered in the update; a general relativistic correction has also been added (Brumberg 1991). Resolving Mercury as a separate body made the largest improvement to the asteroid orbits used as test cases. An additional change to the model concerns the calculation of planet positions. JPL DE406 binary files along with publicly available subroutines making use of Chebychev polynomial interpolation valid from 3000 BCE to 3000 CE are now used. This interpolation scheme is more accurate and allows for the modeling of older streams, as is the case with the Perseids and Aurigids.

2.3 Inputs

Inputs to the model are shown in Table 1. For the Leonids, Perseids, Draconids, and Aurigids, the table lists the parent comet parameters, ejection power law, cap angle of

Table 1 Inputs to the MSFC Meteoroid Stream Model for the Leonid, Perseid, Draconid, and Aurigid meteor showers

Shower	Parent comet	Parent radius (km)	Ejection power law	Cap angle	No. of returns	No. of particles/return
Leonids	Tempel-Tuttle ^a	1.8	$r_h^{-5.0}$	30°	30	300,000
Perseids	Swift-Tuttle ^b	11.0	$r_h^{-6.0}$	60°	9	600,000
Draconids	Giacobini-Zinner ^c	1.7	$r_h^{-0.6}$	30°	28	300,000
Aurigids	Kiess ^d	3.0	$r_h^{-3.0}$	60°	1	550,000

The columns list the parent comet parameters (name and radius), ejection power law as a function of heliocentric distance, cap angle of ejection (an input in the ejection velocity function), the number of returns of the comet considered, and the number of particles ejected per return

^a Radius from Hainaut et al. (1998). Ejection power law determined by fit from previous simulations

^b Radius from Boehnhardt et al. (1996). Ejection power law from Fomenkova et al. (1995)

^c Radius averaged from data given in Newburn and Spinrad (1989), Landaberry et al. (1991), and Churyumov and Rosenbush (1991). Ejection power law deduced from Hanner et al. (1992)

^d No data is available for this comet. Kiess was taken to be an average long period comet and given properties as such—data on long period comets was taken from several sources and averaged

ejection (an input in the ejection velocity function), the number of returns of the comet considered, and the number of particles ejected per return. The comet parameters and ejection power law are determined from the literature. The physical properties of the modeled particles were determined from a uniform, random draw on $\log \beta$, where β is the ratio of radiation pressure forces to the Sun's gravitational force. In the case of each shower, β ranged from $\sim 10^{-5}$ to 10^{-2} , resulting in a mass range between approximately 1 μg and 1 kg, assuming a density of 1000 kg m^{-3} . This particle size range is the range considered a threat to spacecraft.

3 Results and Discussion

An impact parameter (IP) is calculated for each particle approaching Earth within 1 week of the expected shower peak (Moser and Cooke 2004). The particle IPs, in effect the scaled probability that the particle will hit Earth, are summed in 0.005° or 0.01° solar longitude bins, depending on the shower. A Lorentzian is fit to the binned IP versus solar longitude—essentially the flux profile—in order to determine the time of the shower peak.

Figure 1 illustrates how the model update has improved the peak prediction time for the (a) 2001 and (b) 1999 Leonids. It has also improved the predicted duration of the 1999 Leonid storm. The model update does not improve peak prediction time for every stream, however. In the example in Fig. 2, the 1993 Perseids are better constrained by the previous version of the model, both in peak time and duration. It must be noted, however, that in the previous version of the model, 900,000 Perseid particles were simulated for each perihelion passage of the comet, as opposed to the updated model's 600,000 particles per return. As 9 cometary returns were considered, a total of 2.7 million *more* particles were integrated in the previous version. In general, simulating the ejection of more particles from the comet near perihelion yields a greater number of particles intercepted at Earth—and this, in turn, contributes to the overall shape and time of the shower peak. This difference in particle numbers could account for the lack of improvement to the Perseids after the model update, but more work is necessary to determine the culprit.

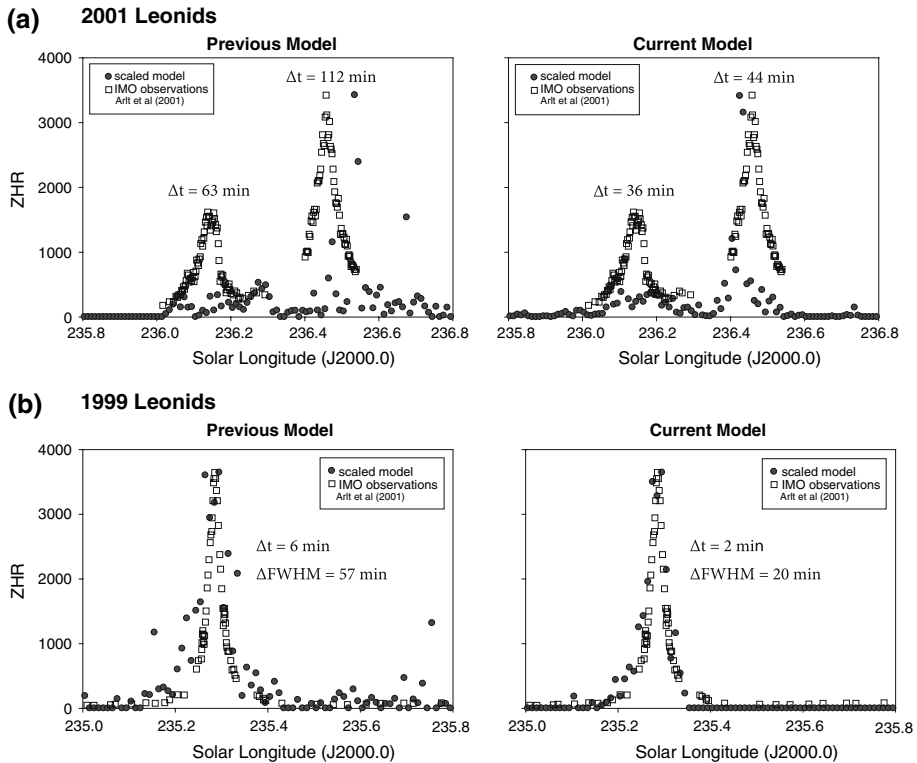


Fig. 1 Comparison of peak prediction times for the (a) 2001 Leonids and (b) 1999 Leonids. Each graph shows the scaled model IPs versus time alongside actual observations. The left panel gives results for the previous version of the model; on the right is the current version of the model. The Δt s listed indicate the difference in time between the model's peak prediction and the observed peak time. The ΔFWHM s in part (b) indicate the difference in full width half maximum between predicted and observed. The current, updated model is an improvement over the previous model

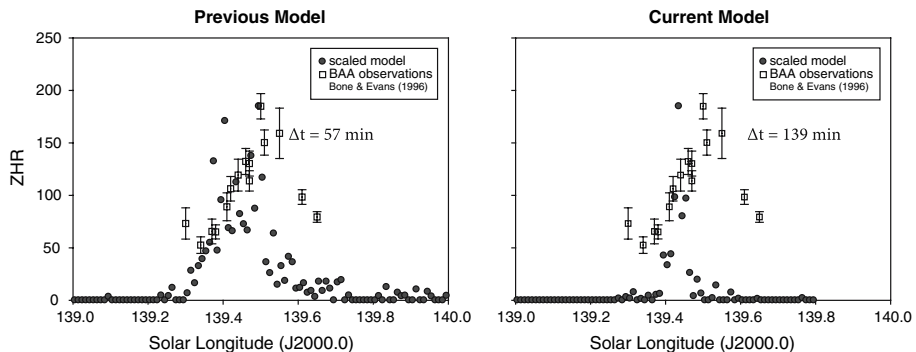


Fig. 2 Comparison of peak prediction times for the 1993 Perseids. See Fig. 1 for an explanation of the graphs. The previous version of the model is better than the current version, although the fact that 2.7 million more particles were studied in the previous version could account for this difference

Figures 3 and 4 show the results of modeling past Draconid storms/outbursts and Aurigid outbursts, respectively. This first attempt at modeling these streams was successful; the 1933, 1946, 1985, and 1998 Draconid peak times were predicted within 1 h of the observed time and the 1935, 1986, and 1994 Aurigid peaks were predicted within 15 min. The Draconid peak predictions can be further refined; the fact that the error is within 1 h is surprising, as the IP approach for the low inclination parent comet 21P/ Giacobini-Zinner was not thought to be valid (Moser and Cooke 2004). The upcoming 2007 Aurigid shower appears similar to the modeled 1935, 1986, and 1994 showers in the number of particles in the vicinity of Earth. It is therefore thought that the 2007 shower will be similar to the past showers in ZHR also: 40–50. It must be noted that the parent body C/ 1911 N1 (Kiesse) is a long period comet and computations of its position are rough estimates at best.

4 Summary

Updates to the MSFC Meteoroid Stream Model better constrain the peak time and duration of the Leonid meteor showers. Improvements to the recent Perseid outbursts were not seen, though this may be accounted for by a failure to run the same number of particles as was done in the previous model version. The MSFC model was put to the task of modeling both

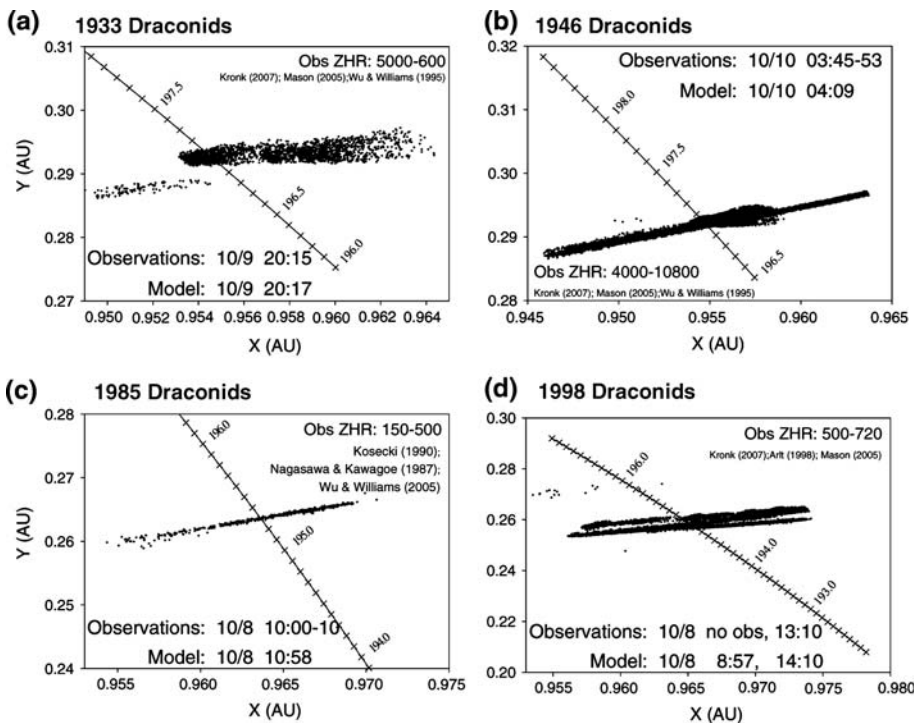


Fig. 3 Recent Draconid outbursts/storms. Each graph is a cross section plot in x-y ecliptic coordinates. The points indicate the nodal crossings of the modeled particles near Earth, represented by the solid line, during the various Draconid outbursts/storms. Peak observed times are listed along with the time the model predicts. Draconid peaks were predicted within 1 hour of the observed peak

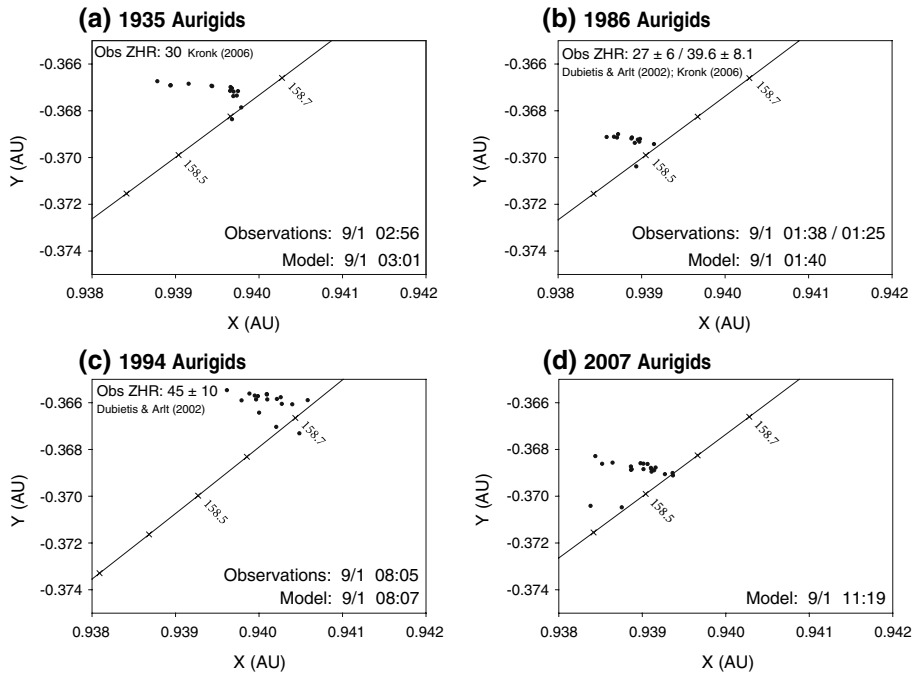


Fig. 4 Recent Aurigid outbursts. See Figure 3 for an explanation of the graphs. Aurigid peaks were predicted within 15 min of the observed peak

the Draconids and Aurigids for the first time this year. The Draconid outburst/storm and Aurigid peak predictions were surprisingly good. There was some concern about the Aurigids this year, but according to the model, the 2007 Aurigids will be on par with showers seen in 1935, 1986, and 1994: ZHR in the 40–50s (no storm).

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